Dmitrii **Altukhov**

Software Engineer · PhD (Computer Science)

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Profile

I'm a software engineer with 12-years experience in designing and building data-processing algorithms and tools. Mathematician by training, I have an academic background, where I built tools and published papers in the area of brain-computer interfaces. It involved real-time processing, analysis and forecasting of brain activity timeseries using both classical methods and deep learning.

Technologies stack Python, PyTorch, Triton, AWS, Kafka, Redis, Java, C++/C, CUDA, SQL, Javascript, React, NodeJS

Mathematical skills Machine learning, digital signal processing, optimization, linear algebra, statistics, algorithms and data structures

Experience

Creative Fabrica — *Amsterdam, the Netherlands*

Python engineer (SupplyAI) — *Dec. 2023 – present*

- Develop, debug and deploy a distributed full-stack generative AI app (**Python**, **Triton**, **React**, **NodeJS**, **AWS**, **Kafka**, **Redis**).
- Redesigned the architecture for image generation backend, which significantly simplified development.
- Developed a color quantization algorithm foundational for one of the best in the industry image vectorization pipelines.

SBDA Group — *Remote*

Data Scientist (contract) — *Sep. 2023 – Oct 2023*

- Designed and implemented an end-to-end algorithmic solution in Python for optimizing a distributed manufacturing process.

Artificial Intelligence Research Institute (AIRI) — *Moscow, Russia*

Research fellow — *Feb. 2022 – Apr. 2023*

- Built and trained deep learning model to predict speech from MEG data in **PyTorch**, meg_speech_decoding on GitHub.
- Developed real-time pipeline for EEG timeseries analysis using **Python**.

Higher School of Economics, Centre for Cognition and Decision Making — *Moscow, Russia*

Junior research fellow — *Feb. 2015 – Dec. 2021*

- Accelerated various parts of my PhD codebase to enable Monte-Carlo statistics, using CUDA (cuBLAS) and C.
- Led 4-people group on a project for real-time feature extraction and GPU-accelerated 3D visualization of brain activity in **Python**
- Sped up by a factor of 10 the beamformer inverse solver in MNE-python by modifying the algorithm for vectorized computations
- Created a pipeline for MEG data preprocessing, see metacognition, MRI_metacognition on GitHub

University of Montreal, CERNEC lab. — *Montreal, Canada*

Visiting researcher, temporary position — *Oct. – Dec. 2015, May 2016 – Dec. 2016*

- Built a classifier for ASD patients vs. Controls with 75% accuracy using classical ML and information geometry in **Python**
- Co-developed an open-source **Python** package for GPU-accelerated brain data processing, Neuropycon, Meunier et al. [2020]

Scientific Research Institute of System Analysis — *Moscow, Russia*

Junior research fellow — *Jun. 2011 – Jan. 2015*

- Developed a droplet evaporation model in **C++**; tested CUDA-accelerated cluster software for simulations of flow in jet engines.

Education

Ph.D. in Computer Science — *Moscow, Russia*

Higher School of Economics, Faculty of Computer Science — *Jan. 2016 – Nov. 2021*

- Thesis: "Optimal methods for functional connectivity estimation in magnetoencephalography."; GPA: 4.0
- Published a paper in a leading neuroscientific journal (see Ossadtchi et al. [2018]) by proposing an algorithm for brain signals analysis.

Specialist degree in Mechanics *(Masters equivalent)* — *Moscow, Russia*

Lomonosov Moscow State University, Department of Mechanics and Mathematics — *Sep. 2008 – Jun. 2013*

- Thesis: "Enhancement and validation of LOGOS software for simulations of the reactive fluid flows."